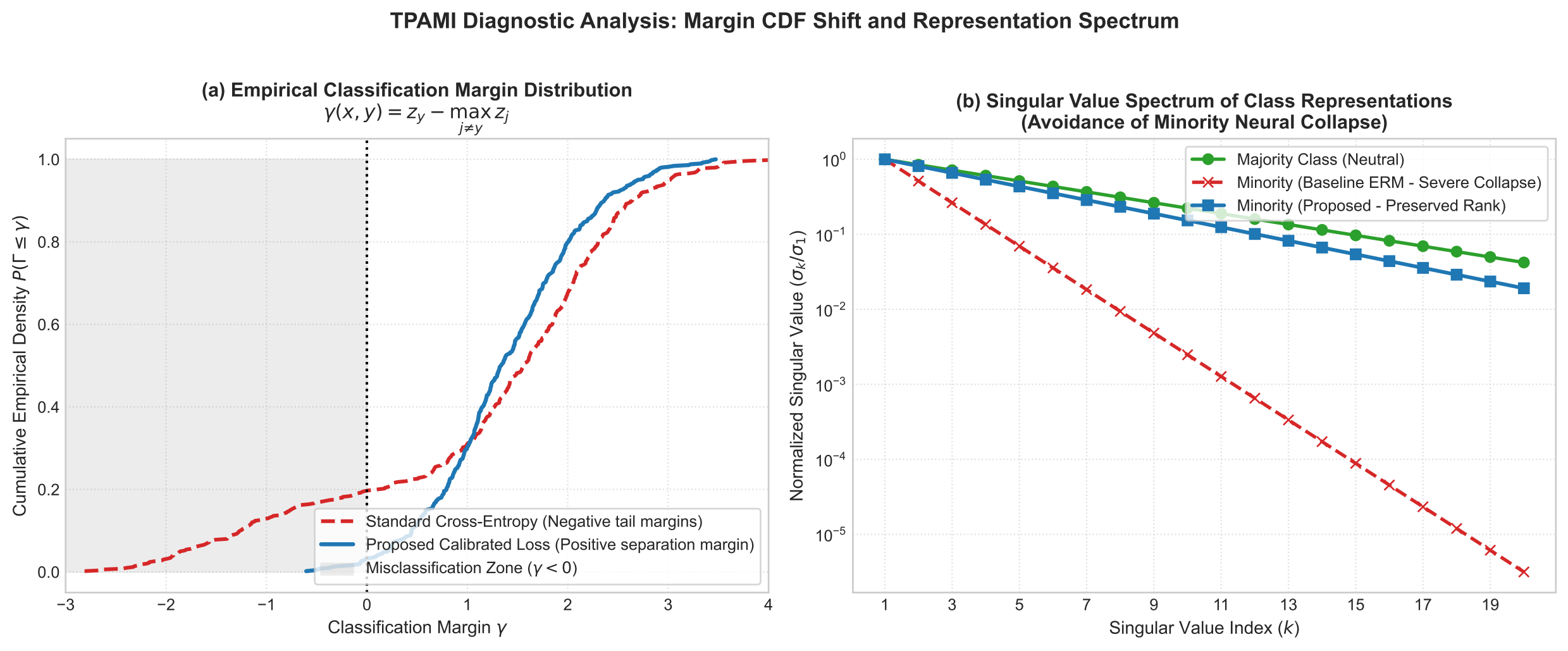
TPAMI Diagnostic Analysis: Margin CDF Shift and Representation Spectrum
(a) Empirical Classification Margin Distribution
$\gamma(x, y) = z_y - \max_{j \neq y} z_j$
Cumulative Empirical Density $P(\Gamma \leq \gamma)$
Classification Margin γ
Standard Cross-Entropy (Negative tail margins)
Proposed Calibrated Loss (Positive separation margin)
Misclassification Zone ($\gamma < 0$)
(b) Singular Value Spectrum of Class Representations
(Avoidance of Minority Neural Collapse)
Normalized Singular Value (σ_k/σ_1)
Singular Value Index (k)
Majority Class (Neutral)
Minority (Baseline ERM - Severe Collapse)
Minority (Proposed - Preserved Rank)